

Haylee Paredes

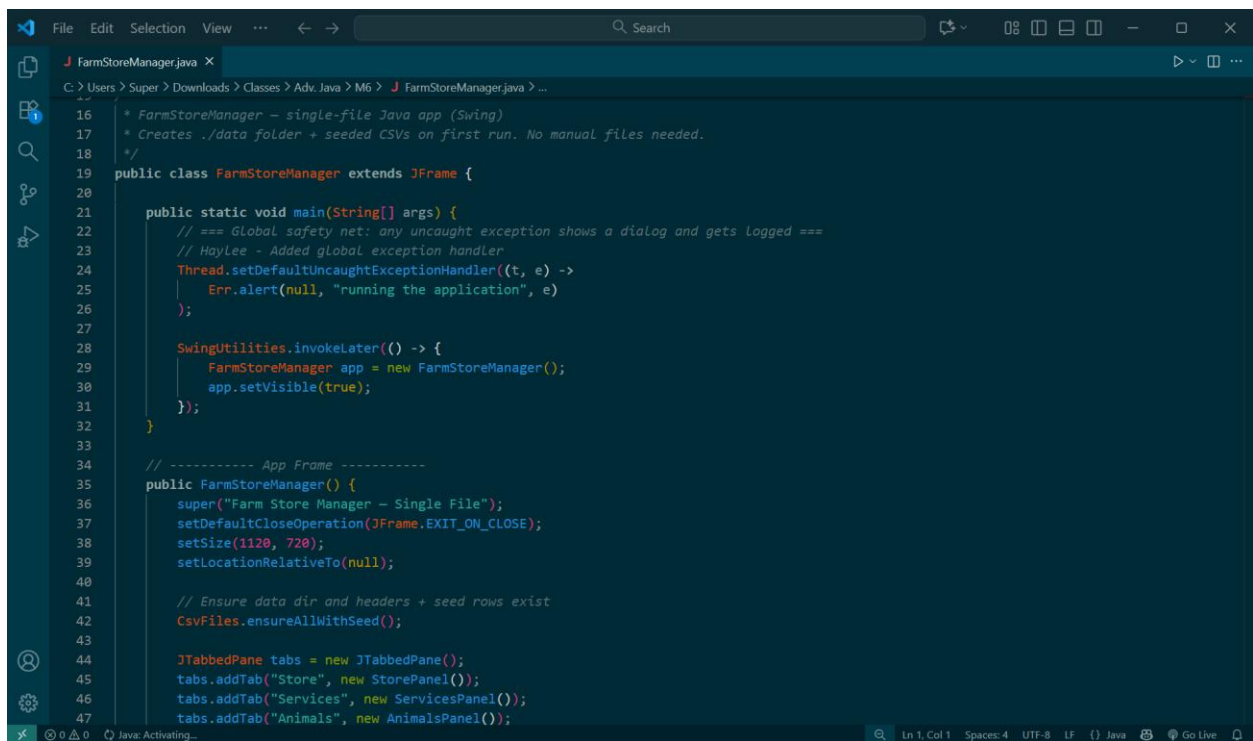
Nov. 12, 2025

CSC-251-0901

Adv. Java Programming

These lines are the new code that I had added which was mostly error handling.

Lines 24-26



```
16  * FarmStoreManager - single-file Java app (Swing)
17  * Creates ./data folder + seeded CSVs on first run. No manual files needed.
18  */
19  public class FarmStoreManager extends JFrame {
20
21      public static void main(String[] args) {
22          // === Global safety net: any uncaught exception shows a dialog and gets logged ===
23          // Haylee - Added global exception handler
24          Thread.setDefaultUncaughtExceptionHandler((t, e) ->
25              Err.alert(null, "running the application", e)
26          );
27
28          SwingUtilities.invokeLater(() -> {
29              FarmStoreManager app = new FarmStoreManager();
30              app.setVisible(true);
31          });
32      }
33
34      // ----- App Frame -----
35      public FarmStoreManager() {
36          super("Farm Store Manager - Single File");
37          setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
38          setSize(1120, 720);
39          setLocationRelativeTo(null);
40
41          // Ensure data dir and headers + seed rows exist
42          CsvFiles.ensureAllWithSeed();
43
44          JTabbedPane tabs = new JTabbedPane();
45          tabs.addTab("Store", new StorePanel());
46          tabs.addTab("Services", new ServicesPanel());
47          tabs.addTab("Animals", new AnimalsPanel());
```

Lines 68-97

```
File Edit Selection View ... Search
J FarmStoreManager.java 1 X
C:\Users\Super\Downloads\Classes>Adv. Java>M6> J FarmStoreManager.java > ...

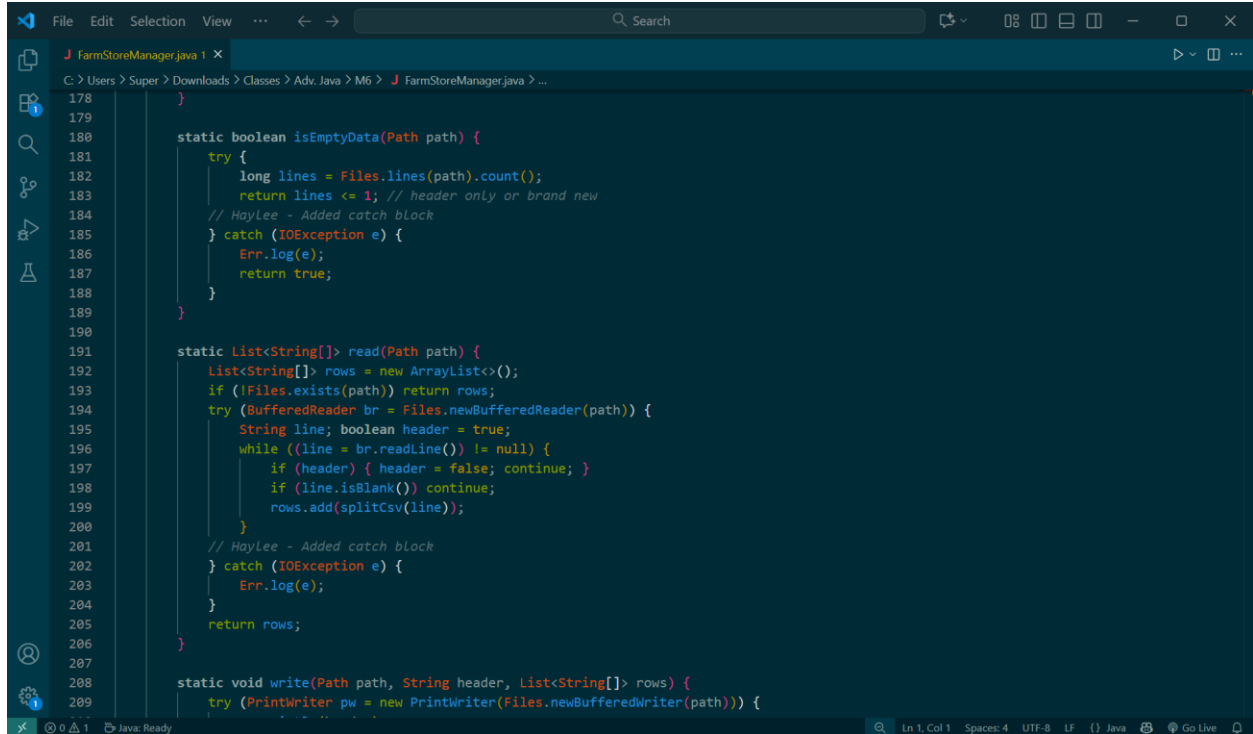
67 // HayLee - Centralized error handling
68 static final class Err {
69     private static Path logPath() {
70         try { Files.createDirectories(Path.of(DATA_DIR)); } catch (IOException ignored) {}
71         return Path.of(DATA_DIR, ...more: "error.log");
72     }
73
74     static void log(Throwable t) {
75         try (PrintWriter pw = new PrintWriter(new FileWriter(logPath().toFile(), append: true))) {
76             pw.println("=== " + LocalDateTime.now() + " ===");
77             t.printStackTrace(pw);
78             pw.println();
79         } catch (IOException ioe) {
80             // Last resort
81             ioe.printStackTrace();
82         }
83     }
84
85     static void alert(java.awt.Component parent, String doing, Throwable t) {
86         log(t);
87         JOptionPane.showMessageDialog(parent,
88             "Sorry, something went wrong while " + doing + ".\nDetails were saved to data/error.log",
89             title: "Unexpected Error", JOptionPane.ERROR_MESSAGE);
90     }
91
92     static Runnable safe(java.awt.Component parent, String doing, Runnable run) {
93         return () -> {
94             try { run.run(); }
95             catch (Throwable t) { alert(parent, doing, t); }
96         };
97     }
98 }
```

Lines 163-165, and lines 174-176

```
File Edit Selection View ... Search
J FarmStoreManager.java 1 X
C:\Users\Super\Downloads\Classes>Adv. Java>M6> J FarmStoreManager.java > ...

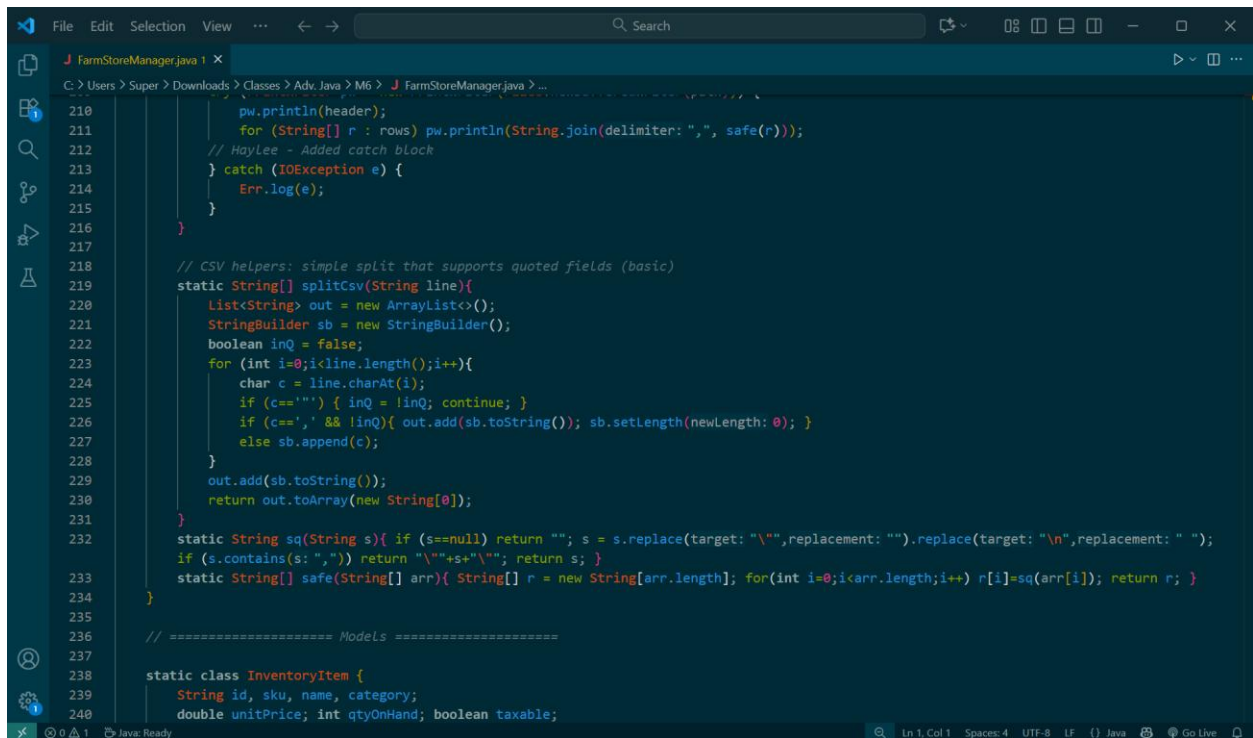
155 }
156
157 static void ensureWithSeed(Path path, String header, List<String[]> seedRows) {
158     if (!Files.exists(path) || isEmptyData(path)) {
159         try (PrintWriter pw = new PrintWriter(Files.newBufferedWriter(path))) {
160             pw.println(header);
161             for (String[] r : seedRows) pw.println(String.join(delimiter: ",", safe(r)));
162             // HayLee - Added catch block
163             } catch (IOException e) {
164                 Err.log(e);
165             }
166         }
167     }
168
169 static void ensureHeaderOnly(Path path, String header) {
170     if (!Files.exists(path) || isEmptyData(path)) {
171         try (PrintWriter pw = new PrintWriter(Files.newBufferedWriter(path))) {
172             pw.println(header);
173             // HayLee - Added catch block
174             } catch (IOException e) {
175                 Err.log(e);
176             }
177         }
178     }
179
180 static boolean isEmptyData(Path path) {
181     try {
182         long lines = Files.lines(path).count();
183         return lines <= 1; // header only or brand new
184         // HayLee - Added catch block
185     } catch (IOException e) {
186         Err.log(e);
187     }
188 }
```

Lines 185-188, and lines 202-204



```
178     }
179
180     static boolean isEmptyData(Path path) {
181         try {
182             long lines = Files.lines(path).count();
183             return lines <= 1; // header only or brand new
184             // HayLee - Added catch block
185         } catch (IOException e) {
186             Err.log(e);
187             return true;
188         }
189     }
190
191     static List<String[]> read(Path path) {
192         List<String[]> rows = new ArrayList<>();
193         if (!Files.exists(path)) return rows;
194         try (BufferedReader br = Files.newBufferedReader(path)) {
195             String line; boolean header = true;
196             while ((line = br.readLine()) != null) {
197                 if (header) { header = false; continue; }
198                 if (line.isBlank()) continue;
199                 rows.add(splitCsv(line));
200             }
201             // HayLee - Added catch block
202         } catch (IOException e) {
203             Err.log(e);
204         }
205         return rows;
206     }
207
208     static void write(Path path, String header, List<String[]> rows) {
209         try (PrintWriter pw = new PrintWriter(Files.newBufferedWriter(path))) {
```

Lines 213-215



```
210         pw.println(header);
211         for (String[] r : rows) pw.println(String.join(delimiter, ", ", safe(r)));
212         // HayLee - Added catch block
213     } catch (IOException e) {
214         Err.log(e);
215     }
216 }
217
218 // CSV helpers: simple split that supports quoted fields (basic)
219 static String[] splitCsv(String line){
220     List<String> out = new ArrayList<>();
221     StringBuilder sb = new StringBuilder();
222     boolean inQ = false;
223     for (int i=0;i<line.length();i++){
224         char c = line.charAt(i);
225         if (c=='\"') { inQ = !inQ; continue; }
226         if (c==',' && !inQ){ out.add(sb.toString()); sb.setLength(newLength: 0); }
227         else sb.append(c);
228     }
229     out.add(sb.toString());
230     return out.toArray(new String[0]);
231 }
232 static String sq(String s){ if (s==null) return ""; s = s.replace(target: "\"",replacement: "").replace(target: "\n",replacement: " ");
233 if (s.contains(s: ",")) return "\""+s+"\""; return s; }
234 static String[] safe(String[] arr){ String[] r = new String[arr.length]; for(int i=0;i<arr.length;i++) r[i]=sq(arr[i]); return r; }
235 }
236
237 // ===== Models =====
238 static class InventoryItem {
239     String id, sku, name, category;
240     double unitPrice; int qtyOnHand; boolean taxable;
```

Lines 567-570

```
J FarmStoreManager.java 1 X
C:\Users\Super\Downloads\Classes>Adv.Java>M6>J FarmStoreManager.java >...

559 JButton add = new JButton(text: "Add Item");
560 JButton edit = new JButton(text: "Edit");
561 JButton del = new JButton(text: "Delete");
562 JButton sell = new JButton(text: "New Sale");
563 actions.add(add); actions.add(edit); actions.add(del); actions.add(sell);
564 add(actions, BorderLayout.NORTH);
565
566 // HayLee - Wrapped action listeners with error handling
567 add.addActionListener(e -> Err.safe(this,doing: "adding an item", this::onAdd).run());
568 edit.addActionListener(e -> Err.safe(this,doing: "editing an item", this::onEdit).run());
569 del.addActionListener(e -> Err.safe(this,doing: "deleting an item", this::onDelete).run());
570 sell.addActionListener(e -> Err.safe(this,doing: "processing a sale", this::onSell).run());
571
572 reload();
573 }
574
575 void reload(){
576     model.setRowCount(rowCount: 0);
577     for (InventoryItem it : InventoryRepo.all()){
578         // HayLee - Changed to show Yes/No instead of True/False
579         model.addRow(new Object[]{it.sku,it.name,it.category,money(it.unitPrice),it.qtyOnHand,yn(it.taxable)});
580     }
581 }
582
583 Optional<InventoryItem> selected(){
584     int r = table.getSelectedRow(); if (r<0) return Optional.empty();
585     String sku = (String) model.getValueAt(r,column: 0);
586     return InventoryRepo.bySku(sku);
587 }
588
589 void onAdd(){
590     String sku = JOptionPane.showInputDialog(this,message: "SKU:"); if (sku==null||sku.isBlank()) return;
```

Lines 679-681

```
J FarmStoreManager.java 1 X
C:\Users\Super\Downloads\Classes>Adv.Java>M6>J FarmStoreManager.java >...

673 JButton donePay = new JButton(text: "Mark Done & Pay");
674 JButton addSvc = new JButton(text: "Add Service");
675 actions.add(newAppt); actions.add(donePay); actions.add(addSvc);
676 add(actions, BorderLayout.NORTH);
677
678 // HayLee - Wrapped action listeners with error handling
679 newAppt.addActionListener(e -> Err.safe(this,doing: "creating an appointment", this::onNewAppt).run());
680 donePay.addActionListener(e -> Err.safe(this,doing: "marking done & collecting payment", this::onDonePay).run());
681 addSvc.addActionListener(e -> Err.safe(this,doing: "adding a service", this::onAddService).run());
682
683 reload();
684 }
685
686 void reload(){
687     model.setRowCount(rowCount: 0);
688     List<Appointment> appts = AppointmentRepo.all();
689     List<Customer> customers = CustomerRepo.all();
690     List<Service> svcs = ServiceRepo.all();
691     for (Appointment a : appts){
692         String cname = customers.stream().filter(c->c.id.equals(a.customerId)).map(c->c.fullName).findFirst().orElse("??");
693         String sname = svcs.stream().filter(s->s.id.equals(a.serviceId)).map(s->s.name).findFirst().orElse("??");
694         model.addRow(new Object[]{a.start, cname, sname, a.status, money(a.paidAmount)});
695     }
696 }
697
698 void onAddService(){
699     String name = JOptionPane.showInputDialog(this,message: "Service name:"); if (name==null||name.isBlank()) return;
700     String desc = JOptionPane.showInputDialog(this,message: "Description:"); if (desc==null) desc="";
701     double price = d(JOptionPane.showInputDialog(this,message: "Base price:"));
702     int mins = i(JOptionPane.showInputDialog(this,message: "Duration (minutes):"));
703     List<Service> all = ServiceRepo.all();
704     all.add(new Service(id: null, name, desc, price, mins));
```

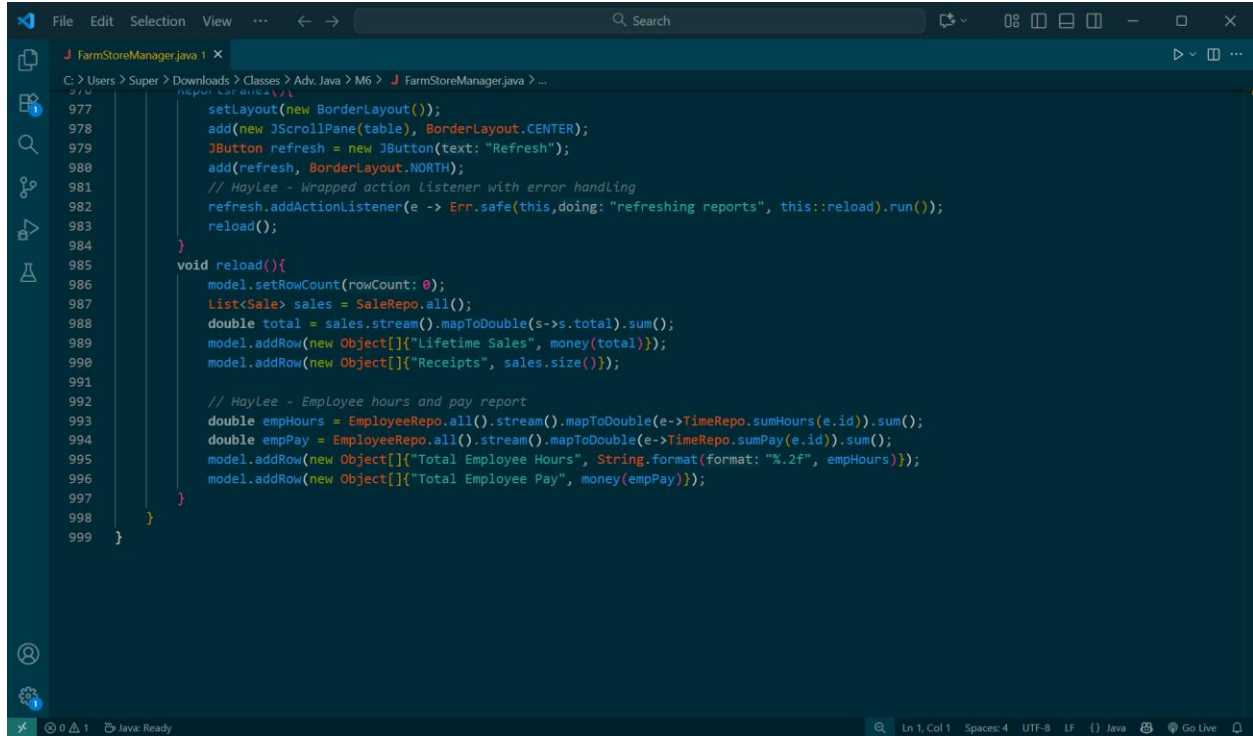
Lines 770-772

```
File Edit Selection View ... Search
J FarmStoreManager.java 1 X
C:\Users\Super\Downloads\Classes>Adv.Java>M6> J FarmStoreManager.java >...
766 JButton sell = new JButton(text: "sell");
767 actions.add(add); actions.add(edit); actions.add(sell);
768 add(actions, BorderLayout.NORTH);
769
770 // HayLee - Wrapped action Listeners with error handling
771 add.addActionListener(e -> Err.safe(this,doing: "adding an animal", this::onAdd).run());
772 edit.addActionListener(e -> Err.safe(this,doing: "editing an animal", this::onEdit).run());
773 sell.addActionListener(e -> Err.safe(this,doing: "selling an animal", this::onSell).run());
774
775 reload();
776
777 void reload(){
778     model.setRowCount(rowCount: 0);
779     for (Animal a : AnimalRepo.all()){
780         // HayLee - Changed to show Yes/No instead of True/False for sold column
781         model.addRow(new Object[]{a.id,a.species,a.breed,a.sex,a.ageMonths,money(a.price),yn(a.onHold),a.supplierName,yn(a.sold)});
782     }
783 }
784
785 Optional<Animal> selected(){
786     int r = table.getSelectedRow(); if (r<0) return Optional.empty();
787     String id = (String) model.getValueAt(r,column: 0);
788     return AnimalRepo.all().stream().filter(a->a.id.equals(id)).findFirst();
789 }
790
791 void onAdd(){
792     String species = JOptionPane.showInputDialog(this,message: "Species:"); if (species==null) return;
793     String breed = JOptionPane.showInputDialog(this,message: "Breed:"); if (breed==null) return;
794     String sex = JOptionPane.showInputDialog(this,message: "Sex (M/F):"); if (sex==null) return;
795     int age = i(JOptionPane.showInputDialog(this,message: "Age (months):"));
796     double price = d(JOptionPane.showInputDialog(this,message: "Price:"));
797     String supplier = JOptionPane.showInputDialog(this,message: "Supplier:"); if (supplier==null) supplier="";
798     AnimalRepo.add(new Animal(id,species,breed,sex,ageMonths,money(price),yn(true),supplier,yn(false)));
799 }
800
801 Java: Ready
```

Lines 860-864

```
File Edit Selection View ... Search
J FarmStoreManager.java 1 X
C:\Users\Super\Downloads\Classes>Adv.Java>M6> J FarmStoreManager.java >...
857 add(actions, BorderLayout.NORTH);
858
859 // HayLee - Wrapped action Listeners with error handling
860 add.addActionListener(e -> Err.safe(this,doing: "adding an employee", this::onAdd).run());
861 edit.addActionListener(e -> Err.safe(this,doing: "editing an employee", this::onEdit).run());
862 toggle.addActionListener(e -> Err.safe(this,doing: "toggling employee active status", this::onToggleActive).run());
863 clockIn.addActionListener(e -> Err.safe(this,doing: "clocking in", this::onClockIn).run());
864 clockOut.addActionListener(e -> Err.safe(this,doing: "clocking out", this::onClockOut).run());
865
866 reload();
867
868
869 void reload() {
870     model.setRowCount(rowCount: 0);
871     List<Employee> emps = EmployeeRepo.all();
872     for (Employee emp : emps) {
873         boolean open = TimeRepo.openShiftFor(emp.id).isPresent();
874         model.addRow(new Object[]{emp.id, emp.name, money(emp.hourlyRate), yn(emp.active), yn(open)});
875     }
876 }
877
878 Optional<Employee> selected() {
879     int r = table.getSelectedRow(); if (r < 0) return Optional.empty();
880     String empId = (String) model.getValueAt(r, column: 0);
881     return EmployeeRepo.byId(empId);
882 }
883
884 void onAdd() {
885     String name = JOptionPane.showInputDialog(this, message: "Employee Name:"); if (name == null || name.isBlank()) return;
886     double rate = d(JOptionPane.showInputDialog(this, message: "Hourly Rate:"));
887     List<Employee> all = EmployeeRepo.all();
888     all.add(new Employee(id: null, name, rate, active: true));
889     EmployeeRepo.saveAll();
890 }
891
892 Java: Ready
```

Line 982



```
File Edit Selection View ... Search
J FarmStoreManager.java 1 x
C:\Users\Super\Downloads\Classes>Adv.Java>M6>J FarmStoreManager.java>...
977     setLayout(new BorderLayout());
978     add(new JScrollPane(table), BorderLayout.CENTER);
979     JButton refresh = new JButton(text: "Refresh");
980     add(refresh, BorderLayout.NORTH);
981     // HayLee - Wrapped action listener with error handling
982     refresh.addActionListener(e -> Err.safe(this,doing: "refreshing reports", this::reload).run());
983     reload();
984 }
985 void reload(){
986     model.setRowCount(rowCount: 0);
987     List<Sale> sales = SaleRepo.all();
988     double total = sales.stream().mapToDouble(s->s.total).sum();
989     model.addRow(new Object[]{"Lifetime Sales", money(total)});
990     model.addRow(new Object[]{"Receipts", sales.size()});
991
992     // HayLee - Employee hours and pay report
993     double empHours = EmployeeRepo.all().stream().mapToDouble(e->TimeRepo.sumHours(e.id)).sum();
994     double empPay = EmployeeRepo.all().stream().mapToDouble(e->TimeRepo.sumPay(e.id)).sum();
995     model.addRow(new Object[]{"Total Employee Hours", String.format(format: "%.2f", empHours)});
996     model.addRow(new Object[]{"Total Employee Pay", money(empPay)});
997 }
998 }
999 }
```

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